

Finally, send the output to the scorer node to measure the performance metric score. Compile and execute all the nodes at each stage of connection to get proper outputs.

Now, for viewing the output as a report, connect it to the PMML (predictive model markup language) writer node. Figure 1 shows the KNIME workflow.

Node configurations

The *CSV reader* node reads CSV files (Figure 2). Use this node if the workflow is created in a server or batch environment, and when the input files structure changes between different invocations. The car evaluation data set is loaded using the *CSV reader* node.

Colours can be assigned for either nominal (possible values have to be available) or numeric columns (with lower and upper bounds). If these bounds are not available, a ‘?’ is provided as a minimum and maximum value. The values are then computed during execution. If a column attribute is selected, the colour can be changed with the colour chooser (Figure 3).

The input table is split into two partitions (i.e., row-wise), e.g., train and test data. In this case, 80 per cent of the data is used for training and 20 per cent for testing (Figure 4).

The *decision tree learner* node induces a classification decision tree in the main memory (Figure 5). The target attribute must be nominal. The other attributes used for decision-making can be either nominal or numerical. Numeric splits are always binary (two outcomes), dividing the domain into two partitions at a given split point. Here the target variable is ‘Evaluation’.

The *decision tree predictor* node uses an existing decision tree (passed in through the model port) to predict the class value for new patterns (Figure 6).

The *scorer* node compares two columns by their attribute-value pairs and shows the confusion matrix, i.e.,

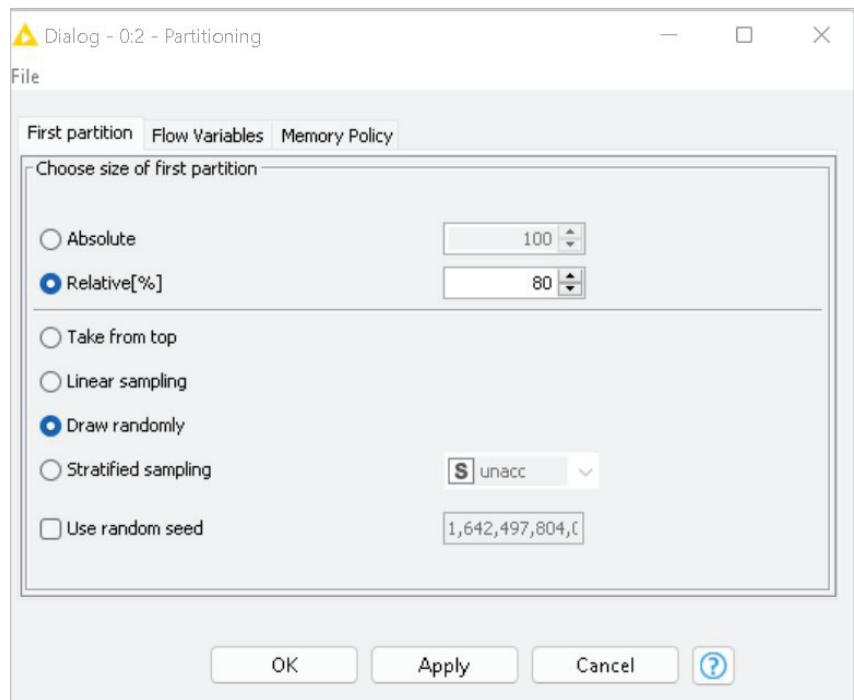


Figure 4: Configuration of *partitioning* node

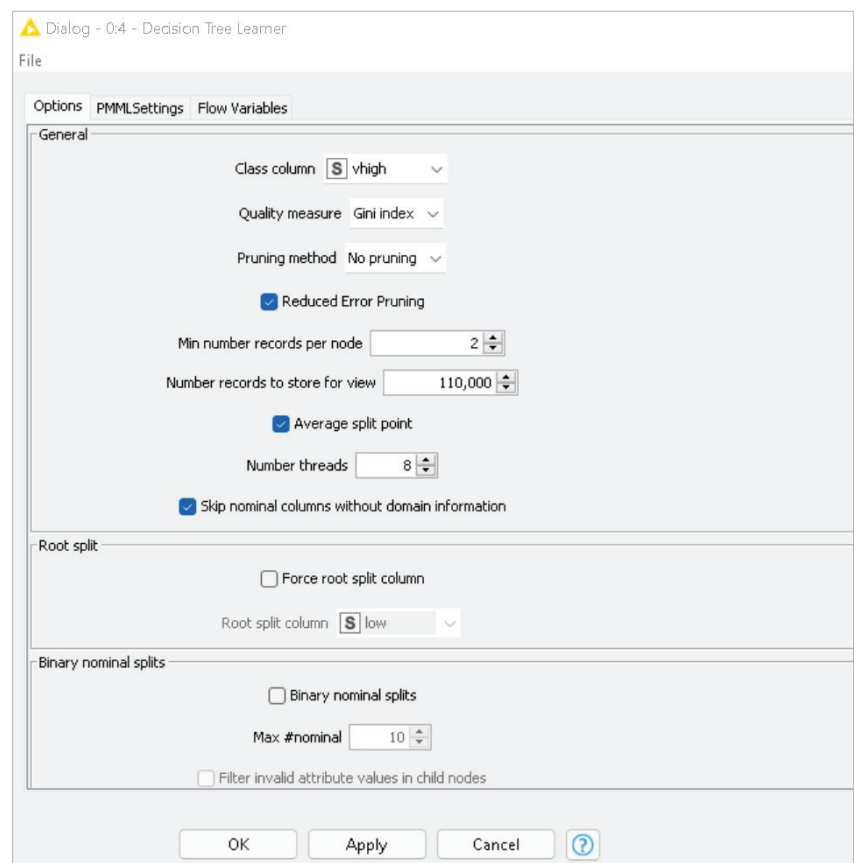


Figure 5: Configuration of *decision tree learner* node